Camera-Based Handwashing Assessment

Visual Recognition, Temporal Segmentation and Adaptive Action Confirmation

Group 43

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Abstract

This project develops a local camera system for recognizing six handwashing actions, reconstructing their order and assessing recognized washing duration. The task requires more than accurate frame labels: hand exchanges, brief occlusion and transitions can introduce spurious steps into an otherwise correct sequence. An initial PSKUS study establishes the value of temporal context and boundary supervision, followed by a self-recorded corpus of 6,987 sampled frames covering standard procedures, reordered actions, omitted steps and visually similar gestures. The proposed pipeline combines YOLO visual features, a unidirectional GRU, learned action boundaries and adaptive confirmation based on segment duration and confidence. Confirmation uses the pace of each recording rather than a fixed three-second admission rule, while preserving observed order. Four pipelines are compared using classification, output boundaries and exact action-chain matching. On 632 aligned valid centers from ten test videos, the proposed model achieves 96.52% accuracy, 93.71% macro F1 and 94.40% boundary F1, with 10/10 final chains matching their references. VideoMAE obtains the highest classification and boundary scores; the proposed model matches its final-chain result with approximately 10.64 million parameters and a recurrent path suited to local feedback. The application supports model selection, incremental prediction, cached finalization, and joint order-and-duration assessment. These results characterize retrospective internal validation in the reviewed scene and support the selected deployment configuration.

Keywords: handwashing recognition; YOLO; GRU; boundary awareness; adaptive confirmation.

1 System Overview

Frame accuracy alone does not guarantee a correct description of a complete washing procedure. Transitions, hand exchanges and occlusion can create brief label errors that become extra steps. Conversely, a recording may contain recognizable gestures while omitting a required step or performing the steps in an incorrect order. Our system therefore combines recognition of the current action with temporal segmentation and confirmation of action instances. It evaluates both what was performed and how the complete sequence unfolded.

Figure 1 summarizes the model. RGB frames are sampled at approximately five frames per second. YOLO extracts visual features, a recurrent encoder tracks their temporal context, and class and boundary heads produce candidate segments. Adaptive confirmation evaluates duration and confidence relative to the recording, then forms an action chain without forcing the standard order. The four-model comparison distinguishes visual recognition, recurrent context, boundary-aware confirmation and video-window modeling.

The interface runs one selected model per session and caches predictions during acquisition. Users receive provisional action and duration feedback while washing; after accepted frames finish processing, the system finalizes the report from this cache. Completion requires the standard six-step order and sufficient recognized washing time, with background intervals excluded. Python and PyTorch implement the backend, with cloud training and local MPS acceleration or CPU fallback. The output describes visible procedure rather than microbiological cleaning effectiveness.

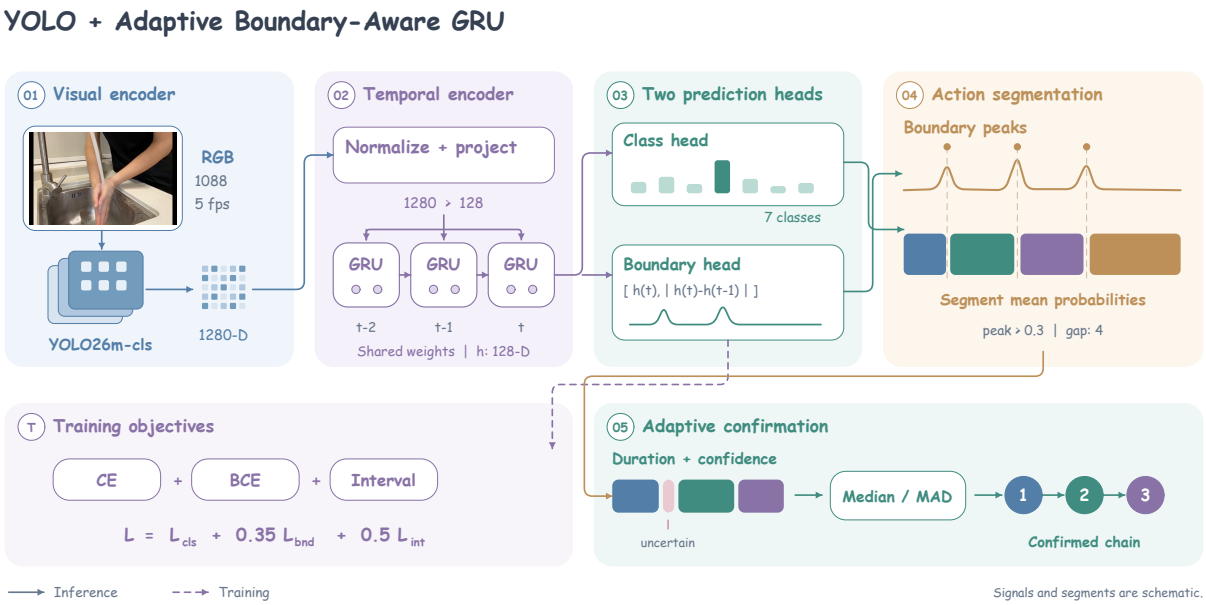


Figure 1: System overview. YOLO features feed a shared recurrent encoder, class and boundary heads, and adaptive action confirmation. The input thumbnail is a real recording; other signals are schematic.

2 Data and Preliminary Evidence

2.1 PSKUS Feasibility Study

The PSKUS resource contains 3,185 annotated handwashing videos from Pauls Stradins Clinical University Hospital [1]. Our preliminary study uses 615 videos from DataSet1–4, split into 430 training, 92 validation and 93 test recordings. Evaluation uses five-frame-per-second sampling, aligned centers of 16-frame windows and a stride of four samples. Agreement between the two designated annotators yields 5,599 evaluable centers, including 1,068 near transitions [4].

Table 1: Preliminary PSKUS comparison, archived on 12 October 2026. All scores are percentages.

Method	Center accuracy	Macro F1	Macro F1 near transitions
YOLO	94.84	90.25	83.27
Initial VideoMAE	89.03	77.21	58.42
Revised VideoMAE	93.39	87.45	79.44
YOLO + GRU	95.98	92.71	90.06
Early boundary-aware YOLO + GRU	96.64	94.06	91.01

Compared with YOLO, ordinary GRU improves macro F1 by 2.46 percentage points and transition-region macro F1 by 6.79 points; early boundary awareness improves both further. Allowing VideoMAE windows to include neighboring actions while keeping the center label stable raises macro F1 from 77.21% to 87.45%. These findings support temporal context and transition-aware training.

Other accounts for 69.7% of valid public-data centers, so macro F1 is reported alongside accuracy. Transition-region macro F1 measures nearby frame labels, whereas the later output-boundary F1 measures detected changes. The early boundary head scores 32.7% under strict adjacent-label-change matching, motivating supervision that accommodates uncertain intervals. Final self-recorded models use neither PSKUS samples nor PSKUS-trained weights.

2.2 Self-Recorded Frames and Annotation

The target-scene corpus contains **6,987 sampled frames with annotation records**, covering standard procedures, reordered or missing actions, and isolated Steps 4 and 5. Continuous annotation retains within-action variation, hand exchanges and transitions. Splits are made by video, keeping adjacent frames from the same recording together [5].

Table 2: Video-level partitions and annotation counts.

Partition	Videos	Sampled frames	Valid class labels	Transition frames
Training	14	2,901	2,822	79
Validation	8	1,364	1,333	31
Test	10	2,722	2,652	70
Total	32	6,987	6,807	180

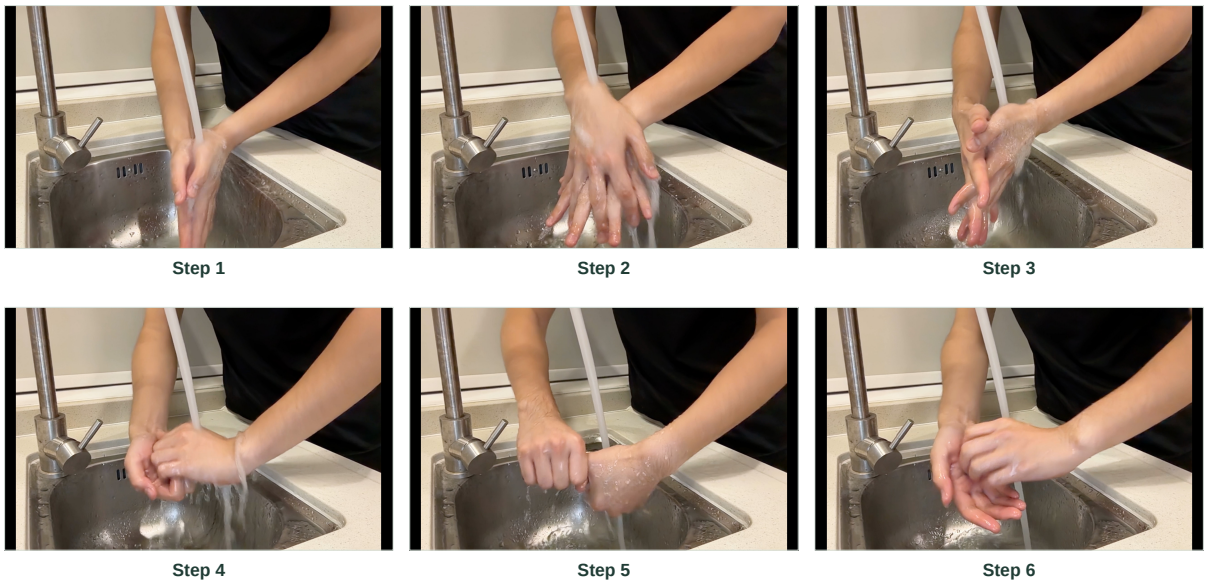


Figure 2: Examples of the six self-recorded handwashing actions, arranged as Steps 1–3 in the upper row and Steps 4–6 in the lower row.

Table 3: Action labels and valid-frame counts across all partitions.

Class	Action	Frames
0	Other / non-step activity	761
1	Palm-to-palm rubbing	729
2	Palm over the back of the opposite hand	1,307
3	Palm-to-palm rubbing with interlaced fingers	709
4	Backs of fingers rubbed against the opposite palm	1,313
5	Rotational rubbing of the thumb	1,130
6	Fingertips rubbed in the opposite palm	858

There are 6,807 definite class labels and 180 uncertain transition frames. Validity flags exclude the latter where supervision or scoring requires a definite class; they are not automatically assigned to Other. Time intervals are transferred to sampled frames and checked against visible hand contact. Video 8 was corrected to chain 124536, with Step 5 starting at 29 seconds.

3 Model and Adaptive Confirmation

3.1 Visual and Temporal Predictions

We compare YOLO, YOLO + GRU, the proposed boundary-aware pipeline, and VideoMAE Base. YOLO uses 1088×1088 frames; both recurrent pipelines reuse its 1,280-dimensional features. VideoMAE uses 16 frames at 224×224 and serves as an accuracy reference. The proposed GRU is unidirectional with 128 hidden units [2, 5].

Let I_t be the input frame, F the frozen YOLO encoder, and $\boldsymbol{\mu}, \boldsymbol{\sigma}$ the feature-normalization statistics. Projection ϕ comprises dropout, a $1280 \to 128$ linear map, GELU and another dropout layer. Class and boundary predictions are

$$\mathbf{f}_t = F(I_t) \tag{1a}$$

$$\mathbf{e}_t = \phi((\mathbf{f}_t - \boldsymbol{\mu}) \oslash \boldsymbol{\sigma}) \tag{1b}$$

$$\mathbf{h}_t = \mathrm{GRU}(\mathbf{e}_t, \mathbf{h}_{t-1}) \tag{2a}$$

$$\mathbf{p}_t = \mathrm{softmax}(W_{\mathrm{c}}\mathbf{h}_t + \mathbf{b}_{\mathrm{c}}) \tag{2b}$$

$$q_t = \mathrm{sigmoid}(g_\theta([\mathbf{h}_t; |\mathbf{h}_t - \mathbf{h}_{t-1}|])) \tag{3}$$

Here $\oslash$ is elementwise division, $\mathbf{p}_t$ contains seven class probabilities, and q_t is a boundary probability. The boundary MLP maps the current state and its absolute change through $256 \to 64 \to 1$ dimensions. Boundary peaks at threshold 0.3, separated by at least four samples, define intervals for class-probability averaging.

3.2 Boundary Supervision

Direct targets require valid labels on both neighboring frames. Uncertain gaps receive interval supervision through the archived objective [5]:

$$\mathcal{L} = \mathcal{L}_{\mathrm{CE}} + 0.35\mathcal{L}_{\mathrm{BCE}} + 0.5\mathcal{L}_{\mathrm{int}} \tag{4}$$

$$\mathcal{L}_{\mathrm{int}} = -\frac{1}{|\mathcal{I}|}\sum_{J\in\mathcal{I}} \log\left(1 - \prod_{t\in J}(1 - q_t)\right) \tag{5}$$

Each $J \in \mathcal{I}$ contains at least one transition. The noisy-OR term rewards boundary evidence within that interval without requiring an exact change frame.

3.3 Adaptive Segment Confirmation

After approximately one-second majority smoothing, adjacent equal labels form segments S_i with duration d_i and mean predicted-class confidence c_i. For non-background indices $\mathcal{A}$, robust relative duration is calculated as

$$m = \mathrm{median}_{i\in\mathcal{A}}(\log d_i) \tag{6}$$

$$s = 1.4826\,\max\{\mathrm{median}_{i\in\mathcal{A}}|\log d_i - m|,\ 0.2\} \tag{7}$$

$$z_i = \frac{\log d_i - m}{s} \tag{8a}$$

$$\widetilde{c} = \mathrm{median}_{i\in\mathcal{A}}\, c_i \tag{8b}$$

With at least four comparable segments, a segment remains unconfirmed when

$$z_i < -2 \quad \wedge \quad c_i < \widetilde{c} \tag{9}$$

With fewer than four, duration rejection is skipped. Historical records lacking confidence can use duration alone; the deployed interface supplies probabilities [5]. The rule replaces a fixed three-second admission threshold, retains observed order and preserves confident short segments. It remains a parameterized statistical procedure, so uncertain short actions can require review.

Algorithm 1 Adaptive boundary-aware action-chain inference

Require: Frames $I_{1:T}$, frozen models, normalization statistics
Ensure: Confirmed action chain $\mathcal{C}$
Extract $\mathbf{f}_t$, update $\mathbf{h}_t$, and predict $\mathbf{p}_t, q_t$.
Select local boundary peaks with $q_t \geq 0.3$ and separation ≥ 4 sampled points.
Decode labels using segment-mean class probabilities.
Apply approximately one-second majority smoothing; form segments S_i.
Record class y_i, duration d_i, and mean class confidence c_i.
Let $\mathcal{A} = \{i : y_i \neq 0\}$; initially mark these segments confirmed.
if $|\mathcal{A}| \geq 4$ **then**
    Calculate $m, s, z_i, \widetilde{c}$ using Eqs. (6)–(8).
    for $i \in \mathcal{A}$ **do**
        if $z_i < -2$ and $c_i < \widetilde{c}$ **then**
            Mark S_i unconfirmed.
        end if
    end for
end if
Collect confirmed non-background labels in temporal order.
Merge adjacent repeated labels to obtain $\mathcal{C}$.
return $\mathcal{C}$

4 Training and Evaluation

YOLO and VideoMAE use the 14-video training and eight-video validation split. VideoMAE starts from the general Kinetics checkpoint `MCG-NJU/videomae-base-finetuned-kinetics`; its sampling rate is 5 fps and window stride is four samples [3, 5]. The temporal heads use frozen YOLO features, AdamW with learning rate 0.001 and weight decay 0.0001, and gradient clipping.

The retained workflow selects epochs and decoder settings on the 14/8 split, then reinitializes and fits the GRU heads and normalization statistics on all 22 development videos. Selected epoch counts are five for ordinary GRU and three for the boundary model. Test videos are excluded. Thus, the experiment compares complete pipelines with their actual fitting and input configurations, rather than an equal-data module ablation [5].

All models share **632 valid scoring centers** across ten test videos [5]. Accuracy measures correct labels; macro F1 equally weights the seven classes. Output-boundary F1 matches predicted label changes one-to-one to reference changes, allowing annotated transition intervals plus 0.4 seconds. Exact-chain match requires the complete sequence, after background removal and duplicate collapse, to equal the reference. A recognition-only summary is

$$S_{\text{recognition}} = 0.4F_{1,\text{macro}} + 0.3M_{\text{chain}} + 0.3F_{1,\text{boundary}} \tag{10}$$

where all terms use the same scale. It excludes size, power and latency. Raw chains use common centers; the proposed final chains additionally use dense 5 fps predictions and adaptive confirmation. Other methods retain their raw chains. Raw-to-final gains therefore reflect the complete pipeline, including sampling density.

5 Results and Error Analysis

Compared with ordinary GRU, the proposed model improves accuracy by 0.48 percentage points, macro F1 by 0.13 points and boundary F1 by 6.81 points. Final-chain matching rises from 6/10

Table 4: Common-center metrics and complete-chain results. Accuracy, F1 and the composite score are percentages; chain columns count exact matches out of ten.

Model	Accuracy	Macro F1	Boundary F1	Raw chain	Final chain	Composite
YOLO	95.25	93.84	76.92	2/10	2/10	66.61
YOLO + GRU	96.04	93.58	87.59	6/10	6/10	81.71
Proposed	96.52	93.71	94.40	8/10	10/10	95.81
VideoMAE	99.53	99.02	97.48	10/10	10/10	98.85

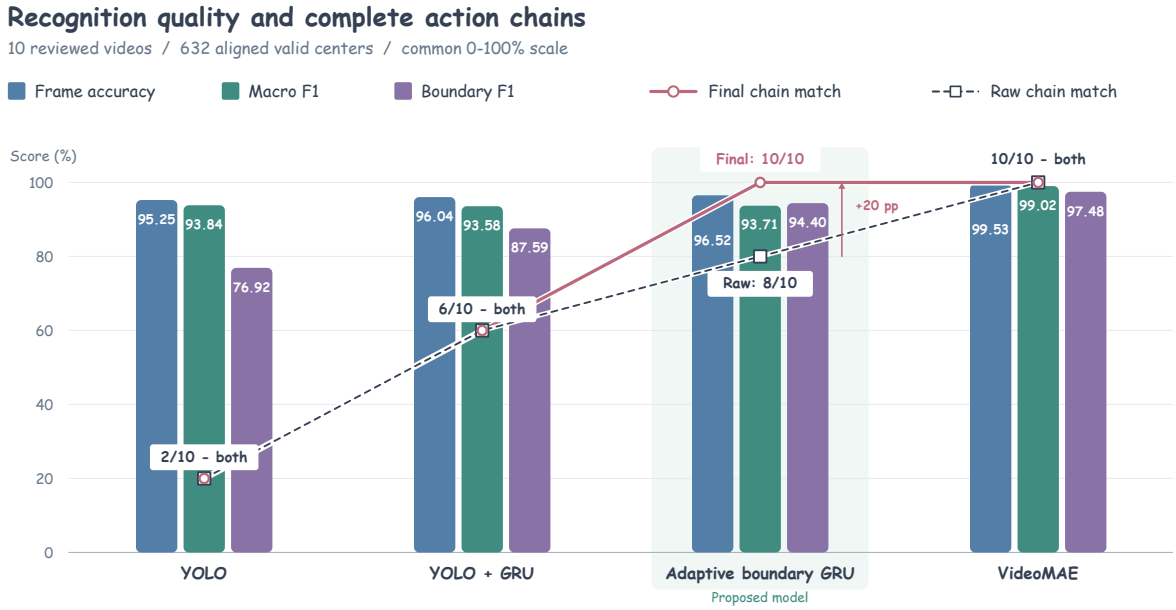


Figure 3: Bars: common-center classification and boundary metrics. Lines: raw and final exact-chain match rates. The proposed final chains use dense 5 fps inference plus adaptive confirmation; the difference is a pipeline comparison, not an isolated ablation.

to 10/10. The larger gains concern segmentation and complete sequences: YOLO already reaches 95.25% frame accuracy but matches only 2/10 chains.

Boundary matching counts (TP/FP/FN) are 60/36/0 for YOLO, 60/17/0 for GRU, 59/6/1 for the proposed model and 58/1/2 for VideoMAE. The proposed model reduces extra boundaries by 64.7% relative to GRU, with one missed boundary. VideoMAE leads classification, boundary F1 and the composite score; the proposed model leads the YOLO family in composite and final-chain results, although its macro F1 is slightly below standalone YOLO.

Matching nonstandard chains and omissions shows that the model describes observed actions rather than enforcing 123456. For original Video 4, ordinary GRU inserts an extra Step 6 (1263456); the proposed final chain is 123456. For original Video 8 and `test_correct_04`, the proposed raw chains 1245346 and 1234562 become 124536 and 123456. These are full-pipeline corrections; removing a step from a chain does not mean every underlying frame is correctly classified.

Step 4/5 confusion must be checked against finger-back contact, thumb enclosure, class mapping and timestamps. Reported scores consistently use the corrected references. An exact reference match describes the recording faithfully; it does not by itself mean the washing procedure follows the required order.

Table 5: Per-video final-chain audit. The proposed prediction matches the reference in every row; digits denote Steps 1–6.

Video	Reference	Proposed prediction
Original Video 4	123456	123456
Original Video 8	124536	124536
`test_correct_01`	123456	123456
`test_correct_02`	123456	123456
`test_correct_03`	123456	123456
`test_correct_04`	123456	123456
`test_error_12356`	12356	12356
`test_error_12564`	12564	12564
`test_error_32654`	32654	32654
`test_error_645321`	645321	645321

6 Camera Application and Deployment

The interface in Figure 4 supports camera and model selection, preview, start and finish controls. Sessions last up to 120 seconds; no audio is collected and recordings stay local. Live predictions are provisional because segment statistics evolve during acquisition. Finalization reuses cached results after any pending frames are processed.

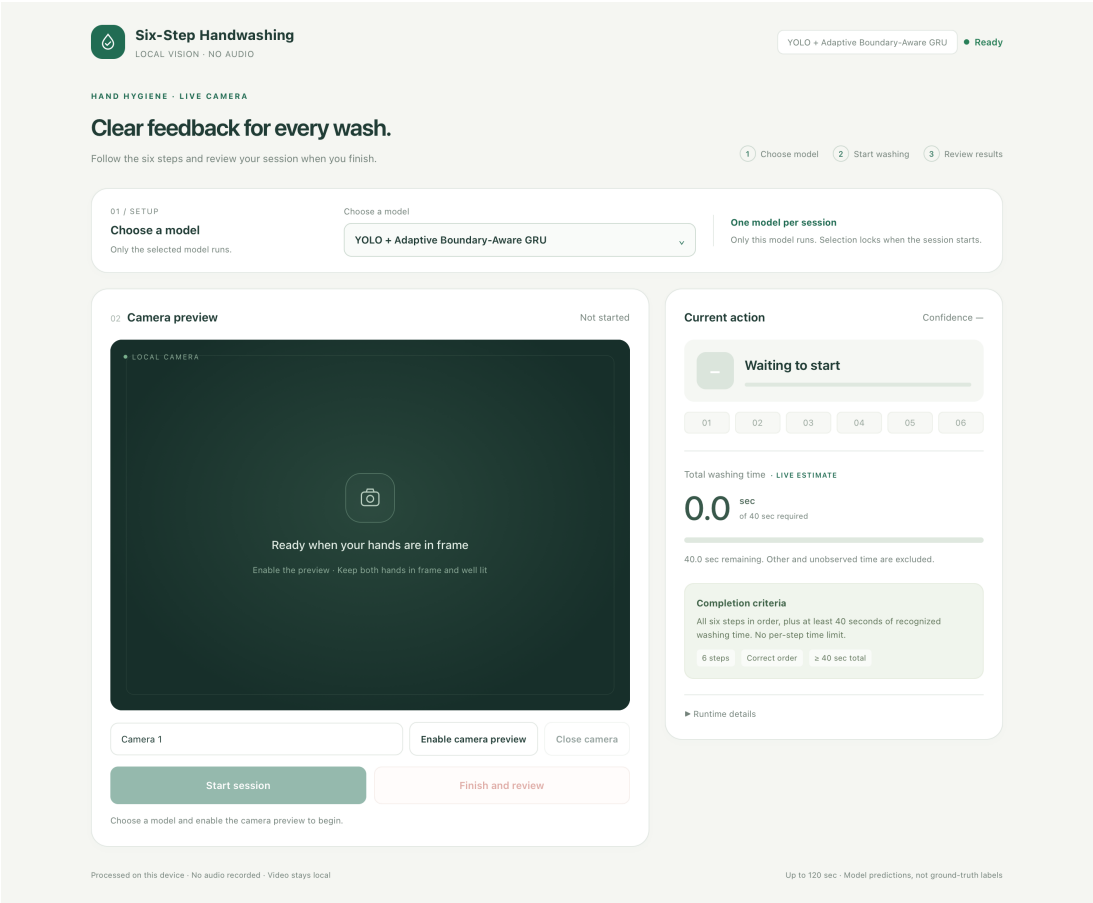


Figure 4: Local camera interface showing model selection, session controls, action feedback and washing-duration assessment before acquisition.

Completion requires chain $1 \to 2 \to 3 \to 4 \to 5 \to 6$ and at least 40 seconds of recognized washing. Other and unknown intervals are excluded; individual steps have no fixed minimum duration. A

brief recognized interval may count toward washing time without becoming a confirmed chain step. The threshold concerns recognized actions, not a full hygiene procedure including wetting, rinsing and drying. Confidence is displayed as model confidence, not accuracy without manual ground truth.

Table 6: Historical local architecture measurements and temporal-input requirements.

Property	Proposed model	VideoMAE Base
Parameters	10,642,639	86,232,583
Inference input	One frame and GRU state	16-frame window
Historical synthetic-input median	148.63 ms per frame and temporal update	317.58 ms per window
Temporal requirements	Recurrent state; confirmation updates as segments accumulate	About 3 s for the first window; about 1.4 s of future context for a centered prediction

VideoMAE has approximately 8.10 times as many parameters. Historical timing uses synthetic inputs on MPS with one warm-up and three measurements; VideoMAE was randomly initialized with the same configuration. A frame/state update and a video window are different units, so these timings do not establish sustained throughput or an end-to-end speed ratio [5].

VideoMAE can stream delayed predictions, but collecting a window and using future context adds waiting time. The proposed recurrent path can classify before collecting 16 frames. Together with fewer parameters and the same final-chain result, this supports its selection for local feedback while retaining VideoMAE as the accuracy reference.

7 Conclusions and Scope

The PSKUS study supports temporal and boundary modeling. On self-recorded data, the proposed pipeline achieves 10/10 final-chain matches while preserving reordered and missing steps. VideoMAE leads frame and boundary metrics; the proposed model offers the selected balance of recognition, model size and local feedback.

These repeatedly reviewed videos, including annotations corrected after prediction inspection, provide retrospective internal validation. Independent evaluation should freeze labels, weights and decoding settings before collecting new users and scenes. Resolution, temporal context and final fitting sets differ across pipelines; sustained throughput and power still require matched hardware measurements.

The delivery includes the application, four frozen models, decoder settings, recordings, 6,987 frame records, split manifests, predictions, metric scripts and checksums. GitHub hosts source and documentation; larger data and weights accompany the full delivery [5].

References

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[5] Practice Project Group (FT) 43. *Self-recorded handwashing system: data, model specifications, training provenance, evaluation and deployment records*. Release `handwash-final-20261022`. Includes `dataset/`, `evaluation/`, `provenance/` and `docs/MODEL_CARD.md`. `https://github.com/Frostash030612/handwash-who-six-step`.